

SQL & PL/SQL Course syllabus

Introduction to Oracle

- RDBMS
- Oracle Architecture
- SGA
- Background processes
- Table spaces, Blocks, Segments
- Execution Plan
- ACID properties
- RMAN overview
- Concurrency control

SQL Topics

- DDL, DML, DCL & TCL statements
- Arithmetic Expression, Concatenation & Literal, Duplicate Rows
- Data types; Comparison, Logical, Range, Pattern & Other operators
- Rules Precedence, Order By Clause, Substitution variables
- Single Row Function & Customized Output
- NVL, NVL2, NULLIF, COALESCE, Decode, CASE
- Reporting Aggregated Data using Group Function
- Windows/Analytic functions
- JOINS - (Inner, Left outer, Right outer, Full outer, Cross)
- Sub-queries SET Operators
- UNION, UNION ALL, INTERSECT, MINUS
- Data Dictionary Views
- Sequence, Synonyms & Indexes
- Views
- Sub-Query
- CTE (Common Table Expression)

Basic PL/SQL overview

- PL/SQL block
- %TYPE, %ROWTYPE
- Exception handling
- Procedures & Functions

Advanced PL/SQL topics

- Exception handling
- Cursors
- Collections
- Bulk processing
- Bind Variables
- Dynamic SQL
- Object types
- Triggers
- Pipelined Table functions
- Autonomous Transactions
- Performance tuning techniques

- Result Cache
- Profiling PL/SQL

Course duration - 27 sessions

- Each session - 1hr 15 mins
- Total hours - 40 hrs
- Includes 2 lab sessions (For student's benefit)
- Recommended frequency (Weekday 1 session & Weekend 1 session). Can be made flexible per student requirement
- Material - PPTs, Code, Recorded videos, Interview questions FAQ, Online help